Progress Report: AI-Generated Code Detection
New Experiments, Methods & Insights

Update for Advisor Meeting — April 2026

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Outline — What's New

1. New SOTA: DeTeCtiveCode (Exp27)
2. CoDET-M4: 22 Methods Explored
3. Full OOD Evaluation (CoDET-M4)
4. Exp_Climb: Data-Efficient Dual-Bench
5. Zero-Shot Suite: 31 No-Training Detectors
6. AICD + DROID Cross-Benchmark Results
7. Consolidated Insights
8. Scale of Work & Updated Evaluation Matrix
9. Next Steps

DeTeCtiveCode — New Best Method (Author F1 = 71.53)

Architecture: HierTree + SupCon + kNN

- **Backbone:** SpectralCode (ModernBERT + AST + Struct + FFT)
- **HierTree:** Family-aware affinity loss (from Exp18)
- **NEW: Multi-level SupCon** on both neural & spectral projection heads (DeTeCtive, NeurIPS 2024)
- **NEW: kNN blend** at test time ($k=32$, $\alpha=0.25$, bank = 50K)

Loss Function

$$\mathcal{L} = \mathcal{L}_{\textbf{task}} + 0.3\mathcal{L}_n + 0.3\mathcal{L}_s + \lambda_{\textbf{hier}}\mathcal{L}_{\textbf{hier}} + \lambda_{\textbf{sc}}\frac{\textbf{SupCon}_n + \textbf{SupCon}_s}{2}$$

Results (CoDET-M4)

	F1	**Δ** vs UniX
UniXcoder (paper)	66.33	—
SpectralCode	69.82	+3.49
HierTreeCode	70.55	+4.22
DeTeCtiveCode	**71.53**	**+5.20**

Per-Class Gains

Human	GPT	Llama	CodeLlama
0.986	**0.780** ↑	0.825	0.745

Nxcode	Qwen1.5	GH source
0.466	**0.490**	**0.576** ↑

CoDET-M4: Full Leaderboard — 22 Methods Ranked

Top-12 by Author 6-Class Macro-F1

#	Method	F1	Key Idea
1	**DeTeCtiveCode**	**71.53**	HierTree+SupCon+kNN
2	HierTreeCode	70.55	Family-tree loss
3	RAGDetect	70.46	Retrieval-augmented
4	**HyperCode**	**70.33**	Hypernetwork head
5	**KANCode**	**70.30**	B-spline KAN head
6	**EnergyCode**	**70.26**	Energy-margin OOD
7	BiScopeCode	70.20	MLM probe stream
8	TTACode	70.20	Test-time adapt
9	MultiGranCode	70.19	CharCNN fusion
10	GroupDRO	70.17	Worst-group opt
11	**WaveCLCode**	**70.16**	Wavelet contrastive
12	SelfDistillCode	70.14	EMA distillation

Yellow = **NEW since last report**

Bottom Methods + Failures

#	Method	F1	Note
13	ProtoCon	70.13	Prototype bank
14	MoECode	70.04	Sparse MoE
15	**TTLCode**	**70.04**	Test-time LoRA
16	**IBCode**	**70.02**	Info bottleneck
17	**MambaCode**	**69.98**	Selective SSM
18	SpectralCode	69.82	FFT spectral
19	GraphStyleCode	69.71	GAT on AST
20	CosineProto	67.80	Cosine proto
—	EAGLECode	62.89	DANN ↓↓

Plateau Finding

Methods #2–17 cluster in **69.8–70.6%**. The gap from HierTree (70.55) to DeTeCtive (71.53) = **+0.98** came from multi-level SupCon + kNN, *not* from architecture changes.

New Methods: Architecture Innovations Explored

KANCode (Exp31) — B-Spline Heads

- Kolmogorov–Arnold Network classifier head
- Learnable B-spline activations (ICLR 2025 Oral)
- EMA teacher auxiliary loss (momentum $= 0.995$)
- Binary: **99.09** (ties best) | Author: **70.30**

MambaCode (Exp36) — SSM Branch

- Selective state-space model (Mamba) branch
- $O(n)$ complexity vs cross-attention $O(n^2)$
- Author: 69.98 — below SpectralCode
- → SSM does *not* help code authorship

HyperCode (Exp32) — Hypernetwork

- Generates classifier weights from token entropy + spectral stats
- Input-conditioned classification head
- Author: **70.33** (+0.03 vs KAN)
- → **Insight**: Both KAN & Hyper hit same Nxcode/Qwen wall ⇒ ceiling is *not* a head capacity issue

Negative Results

- ✗ **IBCode**: Information bottleneck *compresses* style — exactly wrong for attribution (70.02)
- ✗ **MambaCode**: SSM adds nothing (69.98)
- ✗ **CosineProto**: Nxcode→Qwen confusion *worsens* (67.80, 3120/5537 Nxcode misclassified)

CoDET-M4: Complete OOD Evaluation (HierTreeCode)

OOD Source LOO — Table 9 Proxy

Held-out	Macro-F1	Note
CodeForces	58.29	Stable
GitHub	28.34	⇊ catastrophic
LeetCode	**79.63**	Strong
Average	**55.42**	
Paper (UniXcoder)	55.01	+0.41 parity

GH held-out: human recall = **5.71%** — model trained on CF+LC memorizes competitive-programming templates.

✓ First DM method to match UniXcoder on source OOD

OOD-Source avg 55.42 vs paper 55.01 ⇒ **parity achieved (+0.41)**.

OOD Language LOO — Table 10 Proxy

Held-out	Macro-F1	Note
C++	**87.68**	Best
Java	76.92	—
Python	59.86	↓ weakest
Average	**74.82**	
Paper (UniXcoder)	**88.96**	−14.14 gap

OOD Generator LOO

Held-out Gen.	Weighted-F1
CodeLlama	97.59
GPT	87.97
Llama3.1	99.49
Nxcode	99.19
Qwen1.5	99.34
Average	**94.72**

OOD Summary: What We Learned

OOD Results vs Paper

OOD Mode	Paper	Ours	Δ
Generator (W-F1)	—	**94.72**	new
Source (Macro)	55.01	**55.42**	+0.41
Language (Macro)	**88.96**	74.82	−14.14

Key Insights

1. **Source OOD: parity** — first method to match UniXcoder
2. **Language OOD: −14 gap** — Python LOO collapses; lexical shortcuts
3. **GH is universal bottleneck** — both OOD-lang and OOD-source fail on GitHub

GitHub Source = Hardest Distribution

Evidence from *every* experiment:

- OOD-Source held-GH: **28.34%** macro (human recall 5.71%)
- IID per-source: GH = 56.18% vs CF = 77.17%
- ↓21% gap between GH and CF across all methods
- CF/LC = competitive programming (narrow style)
- GH = real-world diverse code (wide style)

Any method breaking 0.40 macro on OOD-SRC-gh is a NeurIPS-level contribution

Exp_Climb: 20% Data, Dual-Benchmark Results

Lean Leaderboard (CoDET + Droid, 20% train)

#	Method	Auth	Py	GH	T3	T4
1	**NTKAlign**	**71.03**	53.3	**35.1**	88.71	88.19
2	**FlowCodeDet**	**70.90**	**64.5**	33.4	89.34	88.30
3	Epiplexity	70.78	63.3	31.0	89.26	**88.46**
4	AvailPredict	70.35	46.5	29.6	88.15	87.31
5	SAMFlat	70.22	54.1	31.4	88.73	87.51
6	Poincaré	69.58	47.6	26.5	**89.76**	87.99
7	PersistHom	70.15	54.4	**35.6**	85.85	87.07
REF	UniXcoder	66.33	—	—	—	—
REF	DroidDet-L	—	—	—	88.78	—

Auth/Py/GH = CoDET Author/OOD-Lang-Py/OOD-Src-GH Macro-F1. T3/T4 = Droid Weighted-F1.

Highlights

- ✓ **NTKAlignCode** 71.03 author with only **20% data**! (vs full-data DeTeCtive 71.53)
- ✓ **FlowCodeDet** OOD-lang-python = **64.50** (+11 pts over pack!)
- ✓ **PoincaréGenealogy** Droid T3 = **89.76** (> DroidDetect-Large)
- ✓ **PersistentHomology** OOD-src-gh = **35.56** (climb record)

Paper Claim

"With only **20%** of training data, our methods *match or exceed* full-data paper baselines on two major benchmarks."

Exp_Climb: Novel Method Highlights

★ FlowCodeDet (Exp_06)

Class-conditioned flow matching auxiliary loss.

- Learns a continuous normalizing flow on the embedding space, conditioned on generator labels
- OOD-lang-python: **64.50** — best by +11 pts
- Droid T3: 89.34, T4: 88.30
- First climb method to clear 70.6 Author

★ NTKAlignCode (Exp_13)

Neural Tangent Kernel alignment on the task head.

- Gram-matrix alignment between NTK and target kernel
- Author: **71.03** (climb #1)
- OOD-src-gh: **35.14** (climb #2)
- Only 0.50 behind DeTeCtive's 71.53 — with 20% data!

★ PoincaréGenealogy (Exp_04)

Hyperbolic embedding in Poincaré ball for hierarchy-aware classification.

- Centering loss organizes Human→AI family tree
- Droid T3: **89.76** (best, > DroidDetect-Large)

★ PersistentHomologyCode (Exp_11)

Topological Data Analysis: Betti numbers from AST filtration.

- OOD-src-gh: **35.56** (climb record!)
- Topology captures structural invariants invisible to tokens

Zero-Shot Detectors: 31 Methods, No Training

Top-10 by Droid T3 Weighted-F1

#	Method	D-T3	CoDET	HR	Signal
1	**BuresQuantum**	**.432**	.422	.881	Quantum
2	CodeAcrostic	.426	.337	.947	Comment
3	CFGEntropy	.414	.364	.948	Control-flow
4	EntropyWM	.360	.409	.955	Cumul. ent.
5	SyntacticPred	.348	.419	.948	Syntactic
6	Martingale	.346	.362	.943	Econometric
7	PathSignature	.342	.335	**.955**	Rough-path
8	SemanticDrift	.340	.351	.945	Paraphrase
9	PIFE	.322	.345	**.952**	Perturbation
REF	*FDG (ours)*	*.321*	*.348*	*.950*	*Curvature*
REF	*FDG (paper)*	*.645*	—	*.840*	—

D-T3 = Droid T3 W-F1. CoDET = Binary Macro-F1. HR = Human Recall (Droid).

FDG Reproduction Gap

Paper Fast-DetectGPT: **64.54** W-F1
Our reproduction: **32.07** W-F1
Gap: $-$**32.33 pts**

Possible cause: paper used full-data access and a larger mask-sampling budget. Our within-suite comparisons remain fair (same protocol).

9 methods beat our FDG reproduction

All 9 top methods $>$ 32.07 on Droid T3.
Best: BuresQuantum +11.17 pts
Best HR $\geq$ 0.95: PathSig + PIFE (3/4 oral claims)

Signal Family Diversity

26 **orthogonal** signal families: curvature, compression, quantum info, path-signature, martingale, optimal transport, attention criticality, . . .
"No single signal dominates code authorship."

Zero-Shot: Novel Signal Families

6 Novel Detectors (Never Applied to Code)

- **PathSignature**: Chen's rough-path iterated integrals on log-prob trajectory. **3/4 oral claims pass.**
- **BuresQuantum**: Attention matrix as quantum density matrix; Bures metric captures "entanglement". **Best W-F1 overall.**
- **Martingale**: De Jong test on AST-depth-conditioned residuals. Econometric efficiency test for authorship.
- **KSDScope**: Kernel Stein Discrepancy over scope-edge graphs. **Only structural (not sequential) ZS detector.**
- **AttentionCriticality**: Hill MLE of power-law exponent on attention avalanches.
- **SinkhornOT**: Entropic optimal transport in embedding space.

Why This Matters

- ✓ **26 orthogonal signal families** — NeurIPS mega-ablation table where reviewer cannot dismiss as "yet-another-log-ratio"
- ✓ **Dual-benchmark evaluation** — every method tested on both Droid T3 & CoDET binary
- ✓ **Human recall $\geq$ 0.95** pinned during calibration — avoids M4/CoDet failure mode
- ✓ Deterministic, reproducible (verified: all 12 metrics bit-identical across 2 runs for Exp_25)

Oral Claim Status

3/4 claims passed: Exp_11 PathSig, Exp_17 PIFE
Pending fix: Exp_15 Bures (HR target after singular-matrix fix)
0/31 beat paper FDG (64.54) — reproduction gap unresolved

AICD + DROID: Deep Methods Suite (6 Completed)

Leaderboard (5-Task Average Macro-F1)

#	Method	AICD T1/T2/T3	DROID T3/T4	Avg
1	SpectralCode	.350	.847	**.549**
2	HierTreeCode	.338	**.849**	.543
3	TokenStat	.330	**.852**	.539
4	MoE-Code	.324	.849	.534
5	SlotCode	.329	.839	.533
6	DomainMix	.286	.716	.458

Per-Task Champions

AICD T1 (hardest):	DomainMix **0.309**
AICD T2 (12-class):	HierTreeCode **0.207**
AICD T3 (4-class):	SlotCode **0.571**
DROID T3:	TokenStat **0.856**
DROID T4:	TokenStat **0.849**

Droid vs Paper (Weighted-F1)

Method	3-cls	Δ
DroidDet-Large	88.78	—
DroidDet-Base	86.76	—
CoDet-M4FT	83.25	—
GPT-SnifferFT	82.75	—
TokenStat (ours)	**89.41**	+0.63
HierTreeCode	89.17	+0.39
SpectralCode	88.77	≈

All 3 methods beat DroidDetect-Base.
TokenStat beats DroidDetect-*Large*.
Using **base** model (149M) vs paper **large** (395M).

AICD T1: Still Unsolved

Val 99.5% → Test 30% for **ALL 23 methods tested**.
OOD collapse is a **dataset property**, not a method bug.

What Works — Validated Patterns

Reusable Components for the Final Method

Pattern	Why It Works	Best Evidence
HierTree family loss	Models generator lineage; cracks Nxcode↔Qwen	Exp18: +3% Qwen F1
Multi-level SupCon	Contrastive on both neural & spectral heads	Exp27: +0.98 over HierTree
kNN blend at test time	Training-free OOD lift	Exp17/27: +0.1–0.3 F1
Token statistics	Cheap, language-agnostic	exp09: best Droid (89.41)
Spectral FFT features	Robust cross-domain transfer	exp11: best AICD avg
Flow matching aux loss	Best OOD-lang-python	Exp_06: 64.50 (+11 pts)
Poincaré embeddings	Hierarchy-aware geometry	Exp_04: Droid T3 89.76

Best cocktail: HierTree + multi-level SupCon + kNN + token stats + spectral FFT
$\Rightarrow$ Current SOTA: DeTeCtiveCode **71.53** Author F1

What Fails — Anti-Patterns to Avoid

✗ Catastrophic Failures

1. **DANN/GRL** (Exp19): Generator-invariant features = *opposite* of attribution. Qwen F1 → 0.198 (random). **−7.66% macro.**
2. **VILW whitening** (Exp05): Whitening loss dominates (∼206), crushes CE capacity.
3. **Orthogonal penalty without warmup** (Exp01): Drives Cov → 0, kills usable features.
4. **IRM without annealing** (Exp06): Penalty explodes to 5000, NaN gradients.

✗ Diminishing Returns

1. **GAT on AST** (Exp23): No gain over BiLSTM (69.71 < 69.82). Needs richer graph (CFG/DFG).
2. **Info Bottleneck** (Exp33): Compresses style = wrong for attribution.
3. **CosineProto** (Exp24): Worsens Nxcode→Qwen confusion (67.80).
4. **Unguarded style contrastive** (Exp16): Division-by-zero → NaN every epoch.

General Rule

Any method that **erases generator-specific style** (DANN, IB, whitening) *hurts* attribution. Style IS the discriminative signal.

The Three Unsolved Bottlenecks

Nxcode ↔ Qwen1.5

33–40% of Qwen samples predicted as Nxcode in *every* method.

Nxcode = fine-tuned from CodeQwen1.5 ⇒ nearly identical "stylometric DNA".

Best so far: Qwen F1 = 0.490 (DeTeCtive).
Target: ≥ 0.55.

HierTree helps +3%, but the family split remains the #1 error source on CoDET-M4.

GitHub Source OOD

OOD-held-GH macro: **28.34%** across all methods.

Human recall = **5.71%** when trained on CF+LC only.

CF/LC = competitive programming (narrow style). GH = real-world (diverse).

Target: > 0.40 (NeurIPS-worthy).
Best climb: PersistHom 35.56.

AICD T1 OOD Collapse

Val 99.5% → Test 29.8% for **ALL 23 methods**.

Not a method bug — it's a **dataset property** (train/test have fundamentally different distributions).

Strategy: Frame as "open challenge / negative result" if reviewers ask.

Best: DomainMix 30.88%.

Scale of Work: By the Numbers

6 methods	22 methods	14 methods	31 methods
Exp_DM **Deep Methods** **(AICD + Droid)**	Exp_CodeDet **CoDET-M4** **(Full data)**	Exp_Climb **Lean Dual-Bench** **(20% data)**	Exp_ZeroShot **No Training** **(Full test set)**
30 designed, 6 run	**22 run, 1 pending**	**14 run, 12 pending**	**20 done, 11 fixing**

73+ unique methods × 3 benchmarks × 3.6M samples

All on Kaggle H100 80GB, BF16, max 12h sessions

Updated Evaluation Matrix

	CoDET-M4 ACL'25	DROID EMNLP'25	AICD EACL'26
Binary	✓ 99.09% (RAG/MoE/KAN)	✓ 89.41% T3	◷ 30.88% T1 (collapse)
Attribution	✓ **71.53%** (6-cls) ↑	—	◷ 20.71% T2
Refined	—	✓ 89.41% T3	◷ 57.06% T3
Adversarial	—	✓ 88.02% T4	◷ T3 (hybrid)
OOD Lang	✓ 74.82% avg ↑	—	◷ 3→9 lang
OOD Source	✓ **55.42%** (parity!) ↑	—	—
OOD Generator	✓ 94.72% W-F1 ↑	—	—
Zero-Shot	✓ 31 methods ↑	✓ 31 methods ↑	—

✓ = completed (with numbers). ◷ = ongoing / challenging. ↑ = new since last report.

Progress since last report:
Author F1: 70.55 → **71.53** (+0.98) | 22 methods on CoDET (was 10) | Full OOD suite done
Lean dual-bench: 14 methods | Zero-shot: 31 methods | Total: 73+ methods

Next Steps & Priorities

High Priority

1. **DFR-SourceBalanced**: Last-layer retrain on balanced data. Target: OOD-GH > 0.40
2. **HierNCoE**: Hierarchical Neural Collapse + ETF. Target: Qwen F1 ≥ 0.50
3. **FrontDoor-NLP**: Causal mediation for OOD-source
4. **Full data training**: Scale 100K $\rightarrow$ 500K/1M

Kill Criteria

A method is worth promoting iff it beats:
• CoDET Author $>$ 71.53 OR • Droid T3 $>$ 89.41 OR
• OOD-SRC-gh $>$ 35.56 OR • AICD T1 $>$ 0.31

Paper Writing

- **NeurIPS 2026** target (deadline: October)
- Headline: "20% data + dual-bench SOTA"
- 3-benchmark evaluation (3.6M samples)
- 73+ method ablation = comprehensive study
- Zero-shot suite = separate appendix / paper

Timeline

Apr–May	Remaining Climb & DM exps
Jun–Jul	Full data scaling + ablations
Aug–Sep	Paper writing + submission
Oct	NeurIPS 2026 deadline

Thank You

Questions & Discussion

CoDET-M4 (ACL'25) · DroidCollection (EMNLP'25) · AICDBench (EACL'26)

73+ methods · 4 experiment suites · 3.6M samples